DIGITAL ELECTRONICS CAPSTONE DESIGN REPORT

Two-Road Traffic-Light Controller using a Moore FSM
and a MOD-6 Synchronous Counter

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Project objective

Design two digital sequential circuits: (1) a two-road traffic-light controller represented as a six-state Moore finite-state machine, and (2) a MOD-6 synchronous counter that counts from 0 through 5 and returns to 0. Both designs use three state bits and a common active clock for their respective registers.

Design scope and evidence note

This report provides the state tables, Boolean next-state equations, K-map layouts, logical block diagrams and verification sequences for both designs. It is a design report, not proof of a completed Logisim Evolution simulation. The supplied Verilog source and testbench are separate implementation aids. Before claiming a Logisim simulation is complete, the flip-flop feedback wiring must be built and run in Logisim, and genuine screenshots/results should be added if the course requires them.

1. TWO-ROAD TRAFFIC-LIGHT CONTROLLER

1.1 Functional description

The controller alternates right-of-way between Road A and Road B. Each road has red, yellow and green outputs. Two all-red clearance states are included between the yellow phase of one road and the green phase of the other road. The controller is a Moore FSM because its outputs depend only on the registered present state.

1.2 State assignment and transition table

State	Q2 Q1 Q0	Road A	Road B	Next state
S0	000	Green	Red	S1 (001)
S1	001	Yellow	Red	S2 (010)
S2	010	Red	Red	S3 (011)
S3	011	Red	Green	S4 (100)
S4	100	Red	Yellow	S5 (101)
S5	101	Red	Red	S0 (000)
Unused	110, 111	Red	Red	S0 (000)

The sequence is S0 → S1 → S2 → S3 → S4 → S5 → S0. The unused encodings 110 and 111 are assigned a recovery transition to 000.

1.3 Next-state Boolean equations

```
D2 = Q0·Q1·Q■2 + Q2·Q■0·Q■1

D1 = Q■2·(Q0 XOR Q1)

D0 = Q■0·(Q■1 + Q■2)
```

Here Q2,Q1,Q0 are the present-state bits and D2,D1,D0 are the inputs to the three D flip-flops. On each active clock edge, the register loads D2D1D0.

1.4 Moore output equations

```
A_GREEN = Q■2·Q■1·Q■0

A_YELLOW = Q■2·Q■1·Q0

A_RED = Q2 + Q1

B_GREEN = Q■2·Q1·Q0

B_YELLOW = Q2·Q■1·Q■0

B_RED = NOT(B_GREEN + B_YELLOW)
```

All outputs are active-high logic signals. For a physical LED circuit, suitable current-limiting resistors and output drivers may be required.

1.5 K-map layouts for next-state logic

Use the Gray-code row order Q2Q1 = 00, 01, 11, 10 and columns Q0 = 0, 1. The 1-cells are listed as minterms using variable order Q2,Q1,Q0.

D2	Q0=0	Q0=1
Q2Q1=00	0	0
Q2Q1=10	0	1
Q2Q1=110	0	0
Q2Q1=100	1	0

D1	Q0=0	Q0=1
Q2Q1=00	0	1
Q2Q1=10	1	0
Q2Q1=110	0	0
Q2Q1=100	0	0

D0	Q0=0	Q0=1
Q2Q1=00	1	0
Q2Q1=10	1	0
Q2Q1=110	0	0
Q2Q1=100	1	0

Minterm forms: D2 = Σm(3,4); D1 = Σm(1,2); D0 = Σm(0,2,4). The simplified equations are shown in Section 1.3. Minterms 6 and 7 are zero because the unused states are deliberately assigned recovery to 000.

1.6 Logical block diagram and implementation

State register: three edge-triggered D flip-flops store Q2,Q1,Q0. Their Q outputs feed the combinational next-state logic. The logic outputs D2,D1,D0 feed the corresponding D inputs. All flip-flops share one clock. Reset initializes the register to 000.

Present-state outputs Q2,Q1,Q0
↓
Combinational next-state logic D2, D1, D0 equations
↓
Three D flip-flops common CLK; RESET = 000
■ Q2,Q1,Q0 feedback to next-state logic
Moore output decoder → A and B signal lamps

A practical controller requires a clock divider or timer to make each state last the desired duration. The six-state FSM by itself advances one state per clock edge; it does not define real-time seconds.

2. MOD-6 SYNCHRONOUS COUNTER

2.1 Functional description

A MOD-6 counter has six valid states and repeats after reaching decimal 5. It uses three D flip-flops because three bits are sufficient to represent six states. The flip-flops share a common clock, so state changes occur synchronously.

2.2 State transition table

Present state Q2Q1Q0	Decimal	Next state D2D1D0	Decimal
000	0	001	1
001	1	010	2
010	2	011	3
011	3	100	4
100	4	101	5
101	5	000	0
110	6 (unused)	000	recovery
111	7 (unused)	000	recovery

2.3 Next-state equations

```
D2 = Q0·Q1·Q■2 + Q2·Q■0·Q■1

D1 = Q■2·(Q0 XOR Q1)

D0 = Q■0·(Q■1 + Q■2)
```

These equations implement the six-state sequence and recover from unused states 110 and 111 to 000 on the next active clock edge.

2.4 K-map layouts

Use the same Gray-code ordering as the traffic-light design. The next-state minterms are D2 = Σm(3,4), D1 = Σm(1,2), and D0 = Σm(0,2,4).

D2	Q0=0	Q0=1
Q2Q1=00	0	0
Q2Q1=10	0	1
Q2Q1=110	0	0
Q2Q1=100	1	0

D1	Q0=0	Q0=1
Q2Q1=00	0	1
Q2Q1=10	1	0
Q2Q1=110	0	0
Q2Q1=100	0	0

D0	Q0=0	Q0=1
Q2Q1=00	1	0
Q2Q1=10	1	0
Q2Q1=110	0	0
Q2Q1=100	1	0

2.5 Terminal-count and wrap-around logic

The terminal count is 101 (decimal 5). Its next state is 000, so the counter wraps around after five. States 110 and 111 also transition to 000 for safe recovery.

2.6 Logical block diagram and implementation

Three D flip-flops: Q2,Q1,Q0 connect to the combinational next-state logic. The logic outputs D2,D1,D0 connect to the respective D inputs. All three clock pins connect to one common clock. An asynchronous reset initializes the counter to 000.

Present-state outputs Q2,Q1,Q0
↓
Next-state combinational logic D2,D1,D0 equations
↓
Three D flip-flops common CLK; RESET = 000
■ Q2,Q1,Q0 feedback to next-state logic

3. VERIFICATION PLAN AND CONCLUSION

3.1 Expected MOD-6 simulation sequence

Clock edge after reset	Q2Q1Q0	Decimal
Reset asserted	000	0
1	001	1
2	010	2
3	011	3
4	100	4
5	101	5
6	000	0
7	001	1

3.2 Expected traffic-light sequence

State	Road A	Road B	Next
S0 (000)	Green	Red	S1
S1 (001)	Yellow	Red	S2
S2 (010)	Red	Red	S3
S3 (011)	Red	Green	S4
S4 (100)	Red	Yellow	S5
S5 (101)	Red	Red	S0

3.3 Simulation procedure

1. Assert reset and confirm the state register is 000.
2. Deassert reset.
3. Apply one clock pulse at a time and observe Q2,Q1,Q0.
4. For the counter, verify 000 → 001 → 010 → 011 → 100 → 101 → 000.
5. For the traffic FSM, verify the six state/output combinations in Section 3.2.
6. Confirm that no state asserts green for both roads simultaneously and that the all-red states occur between directions.

3.4 Conclusion

The two designs use three-bit state registers and combinational next-state logic. The traffic-light controller is a Moore FSM with six defined phases, including all-red clearance states. The MOD-6 counter cycles through six states and wraps to zero after decimal five. The state tables and Boolean equations provide the basis for gate-level or ROM-based implementation and simulation.

3.5 Submission evidence checklist

- ■ Add your name and institution to the cover.
- ■ Attach genuine screenshots of both completed Logisim circuits.
- ■ Include simulation results or waveforms showing the expected sequences.
- ■ Ensure the flip-flop feedback, shared clocks and reset connections are visible.
- ■ Upload the final PDF only after checking the platform's exact capstone requirements.

Important integrity note: this report contains analytical designs and expected verification results. It does not claim that a Logisim circuit was built or simulated. Do not label expected sequences as measured simulation output. If the course requires a live Logisim project link or screenshots, those must be produced from the actual application.

Appendix A — Traffic-light Moore output K-maps

The output functions are derived from the Moore state table. Use Gray-code row order Q2Q1 = 00, 01, 11, 10 and columns Q0 = 0, 1. Minterm numbering uses Q2 as the most significant bit.

A_GREEN = Σm(0) = Q■2·Q■1·Q■0

A_GREEN = Σm(0) = Q■2·Q■1·Q■0	Q0=1
Q2Q1=00	1
Q2Q1=01	0
Q2Q1=11	0
Q2Q1=10	0

A_YELLOW = Σm(1) = Q■2·Q■1·Q0

A_YELLOW = Σm(1) = Q■2·Q■1·Q0	Q0=1
Q2Q1=00	0
Q2Q1=01	0
Q2Q1=11	0
Q2Q1=10	0

A_RED = Σm(2,3,4,5,6,7) = Q2 + Q1

A_RED = Σm(2,3,4,5,6,7) = Q2 + Q1	Q0=1
Q2Q1=00	0
Q2Q1=01	1
Q2Q1=11	1
Q2Q1=10	1

B_GREEN = Σm(3) = Q■2·Q1·Q0

B_GREEN = Σm(3) = Q■2·Q1·Q0	Q0=1
Q2Q1=00	0
Q2Q1=01	0
Q2Q1=11	1
Q2Q1=10	0

B_YELLOW = Σm(4) = Q2·Q■1·Q■0

B_YELLOW = Σm(4) = Q2·Q■1·Q■0	Q0=1
Q2Q1=00	0
Q2Q1=01	0
Q2Q1=11	0
Q2Q1=10	1

B_RED = Σm(0,1,2,5,6,7)

B_RED = Σm(0,1,2,5,6,7) + d	Q0=0	Q0=1
Q2Q1=00	1	1
Q2Q1=01	1	0
Q2Q1=11	1	1
Q2Q1=10	0	1

Output logic note

The unused encodings 110 and 111 are assigned all-red outputs. Therefore A_RED and B_RED remain asserted in those unused states. The K-maps above include these recovery-state output assignments.

Short reflection: A Moore machine was selected because the traffic-light outputs depend only on the registered current state. This makes the output pattern stable between clock edges and separates the lamp-control outputs from direct asynchronous input changes. The trade-off is that a state transition may take one clock edge to affect the outputs; a suitable clock divider is needed to establish practical signal durations.